Who Pays to Compete? Wealth, a Fixed Fee and Entry by Rank

Measurement map

Companion note to the manuscript

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The model is a static, one-shot contest for one prize of value V among Q wealth-heterogeneous challengers and one incumbent. Each challenger chooses $e_i \in \{0,1\}$, paying the fee c to enter; scores $\sigma_i = \mu r_i + (1-\mu)s_i e_i$ are then drawn, and the highest score takes the prize. The realised wealth profile is common knowledge, so the model contains no beliefs, priors or posteriors, and no observer-side parameter such as measurement precision. The discrete objects are Q, the incumbent's label $C \in \{F,G\}$, the decisions e_i, the entrant set S, the count k^*, the index m and the rank j. The continuous objects are wealth, c, V, μ, the draws, the scores and the functions u, κ, F, G, φ, H_m and Δ; the wealth law Λ and the laws of r and s enter as distributions. The utility of wealth u, the prize value V, and every object built from them (κ, Δ in its own units, U_i, and the sums of these across agents) are the modeller's constructs, attributed to the agents to rationalise recorded choices, rather than quantities in the world. The entry decision is a choice between lotteries, because the entry condition is exactly the comparison of the expected payoffs of entering and of not entering. The model is therefore of the von Neumann–Morgenstern kind in the sense of Schoemaker (1982). In a model of that kind utility "is used to represent preferences" rather than determining them (p. 532), and it is unique up to $aU+b$ with $a>0$ (p. 531). It is cardinal in the measurement sense only, since as a preference theory it "provides no more than ordinal rankings of lotteries" and "should not be interpreted as measuring strength of preference under certainty" (p. 533). Sen (1973) draws the matching line on the side of choice. Preference may be defined "as the underlying relation in terms of which individual choices can be explained", and it then "will simply be the binary representation of individual choice" (p. 253). Reading the same relation also as welfare is a further step, and "it is not in general possible to guarantee both simultaneously" (p. 259). Accordingly the entry condition is unchanged under $(u,V) \mapsto (au+b,\, aV)$ with $a>0$, and a statement about these constructs means something only if it survives that transformation. The objects fall into four tables. The players are those whose options the model names, with nature and the institution that sets the rule beside them (the manuscript's table of players). Inputs are fixed before play, model-internal variables are what the model builds from them, and outputs are the choices and outcomes of play. The Observed column codes each object. R is recorded by an observer, C is computed from records, N is not observed and is a fact of the contest or of nature, and K is not observed and is a construct, attributed to the agents to rationalise recorded choices. The last column says how each is obtained. Numerical example parameters (the CRRA coefficient, the witnesses, the constructions in the claim on burden-monotonicity) are not inputs and are omitted, as are the refuted claims R1 and R2 except where the manuscript reuses their objects.

Players

Those whose options the model names, as in the manuscript's table of players, with nature and the institution beside them. Each row says what the player may choose and where that option sits in the tables below.

Table 1: Players and their options.

Symbol	Type	Represents	Observed	Obtained how
challenger i	One of Q agents, $i \in \{0,\dots,Q-1\}$, indexed from the richest.	A player. Its option is to enter or not, $e_i \in \{0,1\}$ (Outputs), and it maximises the one-period expected payoff U_i (Outputs), knowing V, c, μ, the laws of r and s and the whole wealth profile, but not its own draws.	R	Recorded by the observer as a party eligible to enter the contest. Their number is Q (Inputs).
incumbent	One agent, the holder of the position.	A player, scored as an entrant ($C = F$, Inputs). It has no option in M1 to M29, M31 and M32. In M30 its option is to enter or not, paying c like a challenger, as player 0 with the lowest cost.	R	Recorded as the holder before the contest.
nature	Not an agent.	The source of the draws r_i and s_i (Inputs), independent of each other, across players and of wealth. It has no objective.	N	Never observed as such. Its draws enter the recorded scores.
institution	Not a player of the game.	The body that sets the prize, the fee c and the weight μ (Inputs), which the comparative statics move one at a time. Its objective is outside the model. The manuscript's implications C15 to C18 are addressed to it.	R	Recorded through the rules it publishes (the μ row of Inputs).

Inputs

Exogenous objects, fixed before play: the contest, the agents' wealth and utility, the rule, the laws of the draws, and the objects that define a comparative static.

Table 2: Inputs of the model.

Symbol	Type	Represents	Observed	Obtained how
Q	Positive integer.	The number of challengers, "Q challengers plus one incumbent". A fact of the contest.	R	Recorded by the observer as the number of challengers in the contest.
C	A CDF.	The incumbent's score CDF. The incumbent is "scored as an entrant", $C = F$: the open contest, in which the holder competes again under the rule that admits the challengers (manuscript, Appendix D). Rows that hold for any atomless C say so. Her entry is a parameter in M1 to M29 and a choice in M30, where she is player 0 with the lowest cost.	N	Not recorded. $C = F$ is the manuscript's reading of the rule, and in M30 it is the equilibrium outcome whenever identities are pinned. A protected holder, whose C does not move with μ, is outside the model.
w_i; $(w_1,\dots,w_Q)$; $w_{(j)}$, j	Real number, $w_i > c$; a vector in $\mathbb{R}^Q$; its descending order statistics; integer rank, 1 richest.	"Challenger i's resource." The realised profile is common knowledge among the challengers. A fact about each challenger, known to all of them.	R; order statistics C	Recorded by the observer before the entry decisions. Order statistics and ranks are computed from the record. The record is the observer's, whereas the challengers' knowledge of the profile is the manuscript's assumption and is not shown by the record.

Symbol	Type	Represents	Observed	Obtained how
Λ	A distribution across challengers, support bounded below by c.	"The law from which the profile is drawn." A fact of nature.	N	Never observed as a law; each recorded profile is a draw from it. It enters the results only through the realised order statistics, the mean $\bar{\mu}$, the value $1-\Lambda(\bar{\mu})$ and quantile functions.
c	Real number, $c>0$, identical for all agents.	"The nominal cost, identical for all agents", paid "to acquire social capital". In resource units, and what society gives up.	R	Recorded by the observer as the outlay required to enter.
u	Function of wealth, increasing, satisfying burden-monotonicity; a cardinal index, fixed only up to $u \mapsto au+b$, $a>0$, together with $V \mapsto aV$.	"The utility of wealth." The modeller's construct, attributed to the agent, rather than a quantity in the world.	K	Not observed, being a construct. It rationalises the recorded entry decisions, and enters the model only through κ.
V	Real number, $V>0$, in the units of u; scales with a under the transformation above.	The prize's value. The prize is "exogenous and fixed", rank-allocated, a non-market good, additively separable and "valued identically by rich and poor". The modeller's construct, attributed to the agents.	K; the prize itself R	Not observed, being a construct. It enters only as a scale factor of Δ (M1). The prize itself is recorded. Where two prizes are compared (M11), the recorded datum is the agents' choice between them, or their answer to a question ranking them, made before entry. An ordering of V is what rationalises either record. Sen (1973), pp. 257–258, counts answers to questions as information alongside behaviour, noting "well-known limitations of the questionnaire method".
credential	An object, held or not.	The bought component as a thing in the world, the credential that paying the fee buys and the rule reads, Chiba City's paid care or an Italian contracting authority's certificate (manuscript, Appendix D). Its holders are the entrants, the incumbent among them; its price is the fee.	R	Recorded as who holds it, which is the e_i row, and at what price, which is the c row.
published rule	A document.	Closure, in Weber's sense, participation subject to a condition. The condition is holding the credential, the bought component, which paying the fee buys. The rule has two parts the observer can read, the price of the condition, which is the fee c, and how decisive the credential is, which is what it earns against the rest of the index, $1-\mu$ against μ. At one end the credential decides everything and outsiders score nothing; at the other it counts for nothing and paying buys nothing.	R	Recorded as published. Its two parts are read as the c row and the μ row below, the second as an ordering of rules and not a number.

Symbol	Type	Represents	Observed	Obtained how
μ	Real number; the manuscript works with $\mu \in (0,1)$ and the limit $\mu \to 0$.	The weight on the base component r in the score; $1-\mu$ weighs the bought component. A fact of the contest, set by the institution that runs it. The manuscript's Research question motivates it as that institution's trade-off: the bought component is a token it can verify cheaply, where the quality it cares about is costly to verify. The model takes μ as given and does not derive that choice.	N; rules ordered from their published form	Never observed as a number; enters only through F and G. Where the rule is published, the observer reads the value the token earns against the rest of the index (in Chiba City's daycare selection rules, four points for 64 or more hours a month of unlicensed care; in Italian tenders, the extra points each authority sets for a gender-equality certificate; sources in the manuscript's record `lit/applications_2026`), which orders rules by μ without measuring it. Conjecture R1 of the manuscript's Appendix E uses $\sup \operatorname{supp}(G) = \mu$, which holds when the support of r has upper endpoint 1, a condition the primitives do not state.
r_i	Real-valued draw; independent across players, of wealth and of s_i; arbitrary distribution.	The base component: the part of the score every challenger has, whether or not she enters. A fact of nature.	N	Never observed separately. For an outsider the recorded score is μr_i.
s_i	Real-valued draw, $s_i \geq 0$; independent across players, of wealth and of r_i; arbitrary distribution.	The bought component: the part of the score that entry adds. A fact of nature.	N	Never observed separately. For an entrant the recorded score is $\mu r_i + (1-\mu)s_i$.
θ	Real number; $\theta = 0$ is the model.	Tilt of the law of s against wealth rank, outside the model's scope, used in the manuscript's Appendix D to show where count invariance fails (M27).	N	Never observed as a parameter. The independence that θ relaxes is a direct comparison of recorded entrant scores across wealth ranks.
$w \mapsto w'$; T	A map between wealth profiles; T increasing for a pivot-spread.	"Any map between wealth profiles", the change in the wealth distribution that a comparative static is about.	R (two profiles)	Recorded as two wealth profiles, before and after a change, or of two contests. T is seen only at the recorded wealths.
ordering in dispersion	A partial order on wealth laws.	For quantile functions y and z, the second law is more dispersed when $z-y$ is non-decreasing (Lewis and Thompson, 1981); with a crossing point the transport map between the two laws is a pivot-spread, so M13 applies.	C	Computed from two recorded profiles as the sign pattern of the rank differences $w'_{(j)} - w_{(j)}$ along the ranks.
x_0	Real number, in wealth units.	The pivot, "the wealth at which the displacement changes sign."	C	Computed from two recorded profiles. In general it is located only between two adjacent ranks; in the linear family it is given by Recovery 1.
λ	Real number, $\lambda \geq 1$ (M13, M15).	The factor of the linear spread $T_\lambda(w) = x_0 + \lambda(w - x_0)$.	C	Computed from two recorded profiles (Recoveries 1–2).
$\bar{\mu}$	Real number, $\bar{\mu} = \mathbb{E}[w]$.	Mean wealth, and the pivot of the linear mean-preserving spread of M13 (the linear spread).	C	Computed from recorded wealths (Recovery 2).
Λ_α, Λ_β; r_j	Two wealth distributions; quantile levels.	The quantile coupling, which reads the displacement as $\Lambda_\beta^{-1}(r_j) - \Lambda_\alpha^{-1}(r_j)$.	laws N; displacement C	The laws are never observed; the displacement is computed from two recorded sorted profiles. Here r_j is a quantile level rather than the draw r_i.

Model-internal variables

Objects the model constructs from its inputs, which no agent chooses and play does not produce: the cost of the fee, the score laws, the gain from entering and the references built from them.

Table 3: Model-internal variables.

Symbol	Type	Represents	Observed	Obtained how
$\kappa(w) = u(w) - u(w-c)$	Function on (c,∞); strictly decreasing (burden-monotonicity, M23, which strict concavity implies and which is weaker, M24); scales with a under the transformation above. It diverges as $w \downarrow c$ only under the condition of M25, which M15 alone uses.	"The burden of c in utility units", the entry cost (utility). "The sole route by which the wealth distribution affects behaviour." The modeller's construct, attributed to the agent. It is a difference of a utility that represents preferences over lotteries, rather than a measure of how strongly the cost is felt (Schoemaker, 1982, pp. 533, 535).	K	Not observed, being a construct. It rationalises the recorded entry decisions. Only the ratio κ/V survives the transformation, and recorded entry decisions restrict that ratio to a set rather than to a value (Recovery 4).
F, G	CDFs; continuous, common lower endpoint; $F \leq G$ pointwise (M2).	"Law of σ for an entrant (F) and an outsider (G)." Facts of nature. "All results are stated in F and G only."	C	Computed from records as the distributions of recorded scores among entrants and among outsiders, pooled across challengers. Pooling across wealth uses the independence of the draws from wealth (M26 (anonymity)).
$\varphi = G - F$	Function, $\varphi \geq 0$, vanishing at both endpoints of the union of supports.	"The wedge entry buys."	C	Computed only, from F and G.
m	Integer in $\{0,\ldots,Q-1\}$.	The number of other challengers who enter, as faced by a challenger deciding. The rank-j challenger, entering as the j-th entrant, faces $m = j-1$.	C	Computed from the rank.
H_m; M_m	A CDF; a random variable.	The opponent maximum, the "CDF of the best rival score when m other challengers enter". M_m is that best rival score.	H_m C; M_m R	$H_m = F^m G^{Q-1-m} C$ is computed from F, G, C, Q, m. Realisations of M_m are recorded as the best rival score in a contest with m other entrants.
$\Delta(m)$	Real number ≥ 0, in the units of u; a non-increasing function of m (M4 (monotonicity)).	The gain from entering, the right-hand side of the entry condition. A function of the entrant count alone (M26 (anonymity)).	K in its own units; Δ/V C	$\Delta(m)$ itself is a construct, since it carries V. The ratio $\Delta(m)/V = \int H_m\,dF - \int H_m\,dG = \mathbb{E}[\varphi(M_m)]$ (M1) carries no utility and is computed from the recorded F, G, C, Q.
$\Delta(m+1) - \Delta(m)$; K	Real number ≤ 0; $K = F^m G^{Q-2-m} C$, a CDF product.	The step of the gain schedule, equal to $-\frac{V}{2}\int \varphi^2\,dK$ (M3 (the step identity)).	C, as a multiple of V	Computed only, as a multiple of V, from F, G, C, Q.
$\Delta(m) \to V/(m+2)$ as $\mu \to 0$	A function of m, independent of Q.	The M9 benchmark, with the incumbent scored as an entrant. The manuscript's row M9 carries the factor V.	C (a reference)	A computed reference. It is a limit, not reached at any recorded $\mu > 0$.

Symbol	Type	Represents	Observed	Obtained how
$\Delta(m) \leq V/(m+1)$; $k^* \leq V/\kappa$	Bounds, uniform in Q.	M10, saturation.	first C; second K	Statements about constructs. In ratio form, $\Delta(m)/V \leq 1/(m+1)$ is a computed reference from the recorded F, G, C, Q, whereas $k^*\kappa/V \leq 1$ is not observed.
$\kappa(w_{(k^*+1)}) > \Delta(k^*-1)$	An inequality.	The condition of M7 (identities pinned), under which the assortative set is the unique equilibrium.	K	Not observed, being a statement about constructs. In the ratio form $\kappa(w_{(k^*+1)})/V > \Delta(k^*-1)/V$ it survives the transformation of u and V.
d_j; the band	Real numbers; a yes-or-no property of the margin, holding when $d_{k^*} < 0 < d_{k^*+1}$.	$d_j = w'_{(j)} - w_{(j)}$, rank 1 richest. The margin condition is silent exactly inside the band, a pivot-spread never reaches the band, and inside the band the displacement signs determine nothing, for mean-preserving spreads too (M14).	C	Computed from two recorded sorted profiles and the recorded pre-change k^*. Only d_{k^*} and d_{k^*+1} are needed.
$1-\Lambda(\bar{\mu})$	Real number in $[0,1]$.	The entry rate at which the sign of M13 (the linear spread) flips.	C	Computed from recorded wealths, and compared with the recorded entry rate.
$\Delta_Q(0)$; $D(Q) = \Delta_{Q+1}(0) - \Delta_Q(0)$; W_Q	Real numbers; $W_Q = C\,G^{Q-1}(1-G)$, a function.	The first entrant's gain among Q challengers and its step in Q. M16 states that $D(Q) = -V\int W_Q\,(dF - dG)$, for every incumbent law C, and that $\Delta_Q(0)$ rises in Q iff $\int W_Q\,dF < \int W_Q\,dG$.	C (a reference)	A computed reference, from the recorded F, G and Q. It is the change in the gain schedule rather than the recorded change in k^* across contests of different Q.
μ^*	Real number; distribution-dependent.	No row of the manuscript defines it. An earlier draft reported a value of μ at which the sign of $D(Q)$ reverses, found by quadrature, which is an illustration only. The manuscript shows that the sign depends on the law of r (M16 to M22).	not used	Not used.

Outputs

The choices and outcomes of play: who enters, how many, the scores, the winner and the payoffs, with the directions in which the count moves.

Table 4: Outputs of the model.

Symbol	Type	Represents	Observed	Obtained how
e_i	Discrete, $\{0,1\}$.	The challenger's decision to enter, the only choice in the model.	R	Recorded by the observer as whether the challenger paid c, after the profile is known and before the draws.
S	Subset of $\{1,\ldots,Q\}$.	The entrant set, $S = \{i : e_i = 1\}$.	R	Recorded. One play shows the one equilibrium set played, and not the other sets the game admits.
k^*	Integer in $\{0,\ldots,Q\}$.	The entry count, the largest k with the entry condition holding at every rank $j \leq k$. By M6 (count invariance) it is the size of every pure-strategy equilibrium entrant set.	R	Recorded as $\lvert S\rvert$. Its definition runs through κ and Δ, which are constructs, and the recorded count is what that definition rationalises.

Symbol	Type	Represents	Observed	Obtained how
Direction of k^* in V and c	Sign of a weak change.	M11 states that k^* is non-decreasing in V and non-increasing in c.	C (comparison of R)	Direct comparison of recorded k^* across contests that differ only in the recorded c, or only in the prize, with the prizes ordered by the agents' recorded choice between them (the V row of the Inputs table).
Direction of k^* and of Δ in μ	Sign of a weak change.	M28 (headline K8) states that, with the incumbent scored as an entrant, k^* and the gain $\Delta(m)$ at every m are non-increasing in μ, for every atomless law of r and every law of $s \geq 0$.	C (comparison of R)	μ is never recorded (the μ row of the Inputs table), so the comparison is across contests whose rules the observer orders by the weight they give to the base component, with c, the prize and the laws of r and s unchanged. The ordering of the rules is the observer's reading of the rules, not a measurement, and the direction is the only prediction. For Δ, the ratio $\Delta(m)/V$ under each rule is computed from that rule's recorded F and G (Recovery 3), and the two are compared.
Direction of k^* under a change of profile	Sign of a weak change, rise or fall.	M12 (margin condition) and M13 (pivot spreads, the linear spread among them).	C (comparison of R)	Direct comparison of recorded k^* under two recorded profiles, with Q, c, the prize and the score technology unchanged. The hypothesis ($w_{(k^*)}$ against x_0, or the margin condition, which reads d_{k^*} for the rise branch and d_{k^*+1} for the fall branch) is computed from the two profiles and the pre-change k^*. No computed reference is needed.
$k^*(\lambda)$; $\bar{\lambda}$	A step function of λ; a real number.	M15 (strictness) states that k^* is non-increasing in λ, and strictly below $k^*(1)$ for $\lambda > \bar{\lambda}$.	k^* R; $\bar{\lambda}$ K	$k^*(\lambda)$ is recorded only at the profiles observed. $\bar{\lambda}$ is not observed, since the manuscript asserts its existence only, through κ and $\Delta(k^*-1)$.
$w_{(k^*)}$	Real number.	The marginal entrant, the rank-k^* challenger, and her wealth.	C	Computed from the recorded profile and recorded k^* as the wealth of the k^*-th richest challenger, whether or not she is in the recorded S. Under identity multiplicity she can be absent.
$\{(1),\ldots,(k^*)\}$	Subset of size k^*.	The assortative set, the k^* richest, equivalently the k^* with the lowest κ.	C	Computed from the recorded profile and k^*, and compared directly with the recorded S.
σ_i	Real number.	The score, "what the prize is awarded on."	R	Recorded after the draws.
$\mathbf{1}\{i \text{ wins}\}$	Discrete, $\{0,1\}$.	Who takes the prize, the holder of the highest score.	R	Recorded.
U_i	Real number, in the units of u.	The payoff, $u(w_i - ce_i) + V \cdot \mathbf{1}\{i \text{ wins}\}$. The modeller's construct, attributed to the agent.	K	Not observed, being a construct. It rationalises the recorded entry decision. Its level is not fixed by the transformation, since b is free.

Symbol	Type	Represents	Observed	Obtained how
$\sum_i U_i$	Real number, in the units of u.	The unweighted sum of payoffs by which M8 (assortative selection) ranks equilibria.	K	Not observed, being a construct summed across agents, an interpersonal comparison the manuscript's terminology flags (M8). The payoffs summed are those that rationalise the entry choices. Reading their sum as welfare is a second link, which Sen (1973), pp. 254 and 259, says cannot in general be kept together with the first.
$\sum_{i \in S} \kappa(w_i)$; $k^* c$	Real number, in the units of u; real number, resource units.	Total entry cost in utility units, and its resource counterpart, which is identical across equilibria.	first K; $k^* c$ C	The utility total is a construct, and summing it across agents is the interpersonal comparison the manuscript's terminology flags (M8). $k^* c$ is computed from the recorded k^* and c.
Entrants' share of the prize	Real number in $[0,1]$.	M31: the probability $\xi(k^*)$ that the prize goes to an entrant, the incumbent included; non-decreasing in the count, $\frac{k+1}{Q+1}$ with k entrants when entry buys nothing, and non-increasing in μ.	R (frequency)	Recorded directly: whether the winner paid c, across contests run under one rule. Its frequency estimates $\xi(k^*)$. Against the benchmark $\frac{k+1}{Q+1}$, its value when the bought component adds nothing, a recorded share above the benchmark is evidence that the bought component counts, and a share at it is not (M31, M33). No construct enters, unlike every quantity built on u.
r_W	Real number.	The winner's base component. M32 states that it does not fall in μ with the entrants fixed, pointwise, and that once entry responds its expectation can fall, as with a common bonus, a bought component equal for every holder.	N	Never observed, since r is not recorded separately from the score. For $s \geq 0$ the recorded score bounds it from above, $r_W \leq \sigma_W / \mu$; across rules the direction is a prediction of the model, not a recorded comparison.

Structure

- *Entering only through others.* u enters only through κ; wealth enters only through $\kappa(w_{(j)})$ (M26 (anonymity)); μ and the laws of r and s enter only through F and G; V enters only as a scale factor of Δ, and the entry condition is unchanged under $(u,V) \mapsto (au+b, aV)$, $a > 0$; Λ enters only through the realised profile, $\bar{\mu}$, $1-\Lambda(\bar{\mu})$ and quantile functions.
- *Distributions.* Λ across challengers, and the laws of r and s, which reach the results only as F and G.
- *Needing a computed reference.* Δ/V and its step, the M9 benchmark, the first M10 bound in ratio form, the M16 sign, μ^*, and the flip entry rate $1-\Lambda(\bar{\mu})$.
- *Given directly by recorded comparisons.* $F \leq G$ (recorded scores of entrants against outsiders); independence of the draws from wealth (recorded scores across wealth ranks, within one entry status); the recorded S against the assortative set; the direction of k^* across contests differing only in c; the direction of k^* across contests whose rules the observer orders by the weight on the base component; the direction of k^* across two recorded profiles, with its hypothesis checked on the same records; the entrants' share, from whether each winner paid c.
- *Facts that exist but are never observed.* μ, r_i, s_i, and Λ as a law.
- *Constructs, not observed.* u, κ, V, Δ in its own units, U_i, $\sum_i U_i$, $\sum_{i \in S} \kappa(w_i)$, the condition of M7 (identities pinned), the second M10 bound, and $\bar{\lambda}$. Each rationalises recorded entry decisions, or a recorded choice

between prizes. Under $(u,V) \mapsto (au+b, aV)$ the quantities that do not change are exactly the functions of κ/V (and of Δ/V, which carries no utility). U_i and $\sum_i U_i$ move with a and b, whereas $\sum_{i\in S} \kappa(w_i)$ moves with a only. The difference of either sum between two equilibria keeps its sign. Summing across agents is the interpersonal comparison the manuscript's terminology flags (M8).

Recovery formulas

1. *Linear family* $T_\lambda(w) = x_0 + \lambda(w - x_0)$. From two recorded sorted profiles, take ranks a, b with $d_a = w'_{(a)} - w_{(a)}$ and $d_b = w'_{(b)} - w_{(b)}$. Then

$$\lambda = 1 + \frac{d_a - d_b}{w_{(a)} - w_{(b)}}, \qquad x_0 = \frac{d_b\, w_{(a)} - d_a\, w_{(b)}}{d_b - d_a}.$$

   The formula for λ needs $w_{(a)} \neq w_{(b)}$. The formula for x_0 needs $d_a \neq d_b$, equivalently $\lambda \neq 1$ and $w_{(a)} \neq w_{(b)}$. Both use four recorded wealths.

2. *Linear mean-preserving spread of M13 (the linear spread).* $\bar{\mu}$ is the mean of the recorded profile, and the spread about $\bar{\mu}$ preserves that mean. Then $\lambda = (w'_{(j)} - \bar{\mu})/(w_{(j)} - \bar{\mu})$ for any rank j with $w_{(j)} \neq \bar{\mu}$. The formula uses one recorded profile and one rank of the other.

3. *Gain schedule, up to* V. $\Delta(m)/V = \int H_m\, dF - \int H_m\, dG = \mathbb{E}[\varphi(M_m)]$, from the recorded F, G, C and Q (M1). The formula needs the regularity of the primitives (F, G continuous, common lower endpoint).

4. *Ratios that rationalise a recorded equilibrium, as a set rather than a value.* For a recorded equilibrium S with $|S| = k$, $\kappa(w_i)/V \leq \Delta(k-1)/V$ for $i \in S$ (when $k \geq 1$), and $\kappa(w_i)/V > \Delta(k)/V$ for $i \notin S$ (when $k \leq Q-1$). These are the equilibrium conditions of M6 (count invariance) divided by $V > 0$, with Δ/V from Recovery 3.

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